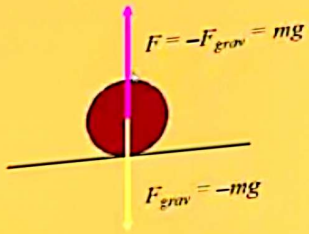
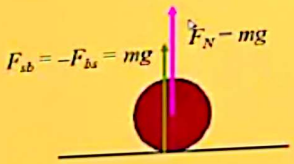
Gravity at Rest

- The force of gravity acts on all objects all the time.
- If an object is at rest, the law of inertia says that the net force is zero.
- There must be a force opposite to gravity that cancels it out.



Normal Force



- The force that opposes gravity for objects on the ground is called the *normal force*.
- It is perpendicular (normal) to the plane of the ground.
- The force is a result of the law of reaction.

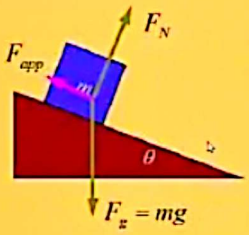
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Normal Force and Weight

- The normal force pushing up against gravity can be measured.
- We measure weight with a scale that measures normal force.
- Weight is a force, not a mass.
- Pounds measure weight, so force (not mass) can be measured in pounds.



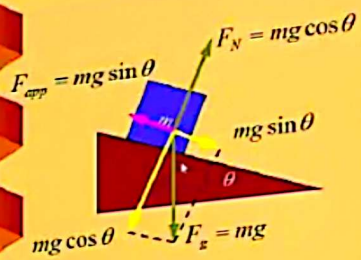
Forces on a Slope



- Draw the forces acting on the block.
  - The force of gravity points down with magnitude $F_g=mg$.
  - The normal force F_N points away from the surface of the inclined plane.
  - An applied force F_{app} holds the block in place.

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Net Force

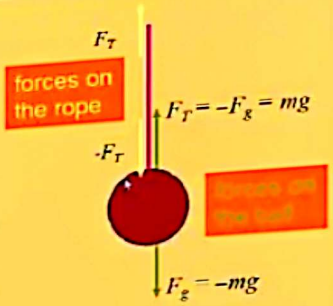


- The block stays in place on the plane.
  - No net force along the surface
  - No net force away from the surface
- Find the components of gravity in those directions.
- Find the unknown forces

The diagram is the same for constant velocity!

Tension Force

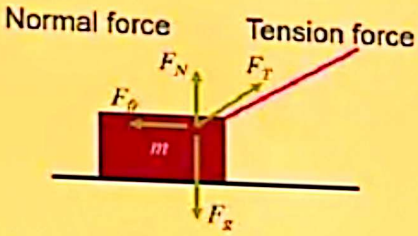
- A taut rope has a force exerted on it.
- If the rope is lightweight and flexible the force is uniform over the entire length.
- This force is called *tension* and points along the rope.
- Tension acts on both ends.



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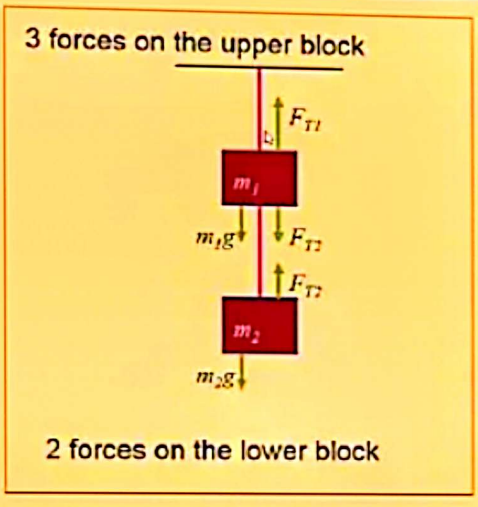
Tension or Normal Force

- Tension and normal forces are different.
  - A pull on an object – tension
  - A push from a surface – normal force
- Either one or both may be present.



Equilibrium in One Dimension

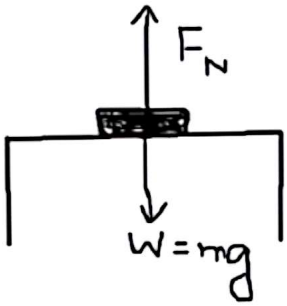
- Two weights are hung supported by strings.
  - On the lower block the two forces balance: $F_{T2} = m_2 g$
  - On the upper block there are three: $F_{T1} = m_1 g + F_{T2}$
  - $F_{T1} = (m_1 + m_2)g$
- The upper string has more tension than the lower string.



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Example

Calculate the normal contact force when a book of mas 3kg is placed on a horizontal table.



Sol

$$m = 3\ kg$$

$$W = mg$$

$$= 3(10)$$

$$= 30N$$

$$F_N = W$$

$$= 30N$$

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Example
Two smooth boxes A and B of masses 20kg and 30kg respectively are placed as shown in the diagram. Find the magnitude of the forces acting on each box.



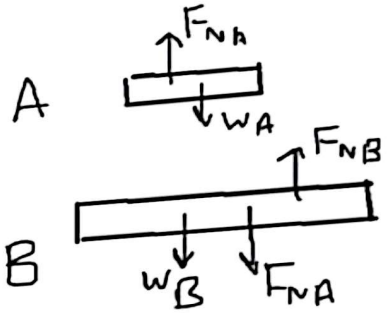
Sol

Box A $m_A = 20kg$

$W_A = mg$

$= 20(10) = 200N$

$F_{NA} = W_A = 200N$



Box B $m_B = 30kg$

$W_A = mg$

$= 20(10) = 200N$

$F_{NA} = W_A = 200N$

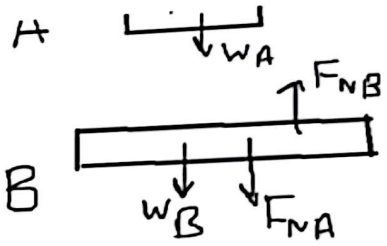
Box B $m_B = 30kg$

$W_B = 30(10)$

$= 300\ N$

$F_{NB} = 300 + 200$

$= 500\ N$



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Three identical smooth balls each of mass mkg are labelled as A, B and C and they rest in contact with each other in a smooth cylinderical container. Write down, in terms of m and g:

(i) The magnitude of the force exerted on ball A by ball B

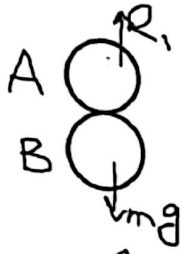
(ii) The magnitude of the force exerted on ball B by ball C.

(iii) The magnitude and direction of the resultant force exerted by ball A and C on ball B.



Sol

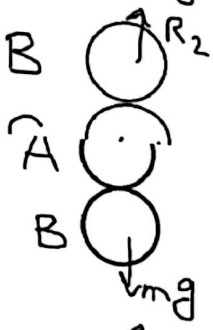
(i)



exerted by ball A and C on ball B.

Sol

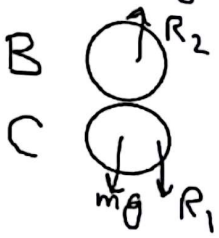
(i) $R_1 = mg$



(ii) $R_2 = mg + R_1$

$= mg + mg$

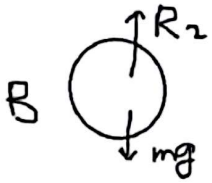
$= 2mg$



(iii) $R = R_2 - mg$

$= 2mg - mg$

$= mg$ act upward

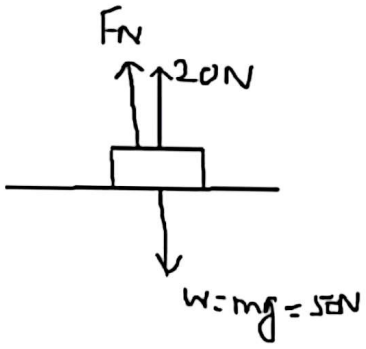


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Example

A box of mass 5 kg is placed on a smooth table. A force of 20N is acting on the box as shown in the figure. Find the contact force acting on the box.



Sol:-

$m = 5\ kg$

$W = mg$

$= 5(10)$

$= 50\ N$

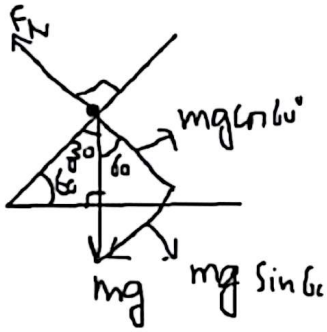
$F_N + 20 = 50$

$F_N = 50 - 20$

$= 30\ N$

Example

A particle of mass 4kg is placed on an inclined plane whose inclination with the horizontal is 60degrees. The plane is smooth. Find the contact force acting on the particle.



Sol:-

$F_N = mg\ Cos 60^\circ$

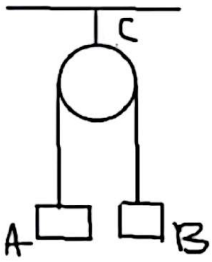
$= 4(10)\ \frac{1}{2}$

$= 20\ N$

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Example
Two particles of mass 2kg each are connected by a light inextensible string pasing over a smooth fixed pulley, which is attached to a string C as shown in the fig. the particles are at rest. Find the tension in the string C.

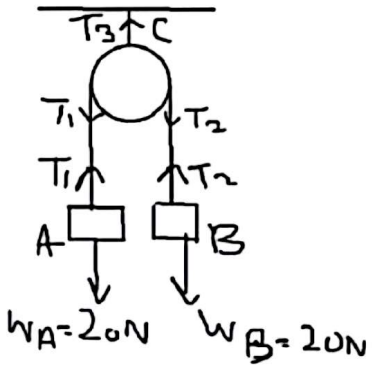
Sol:-



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Test: Find the tension in the string C.



Sol:-

$m_A = 2\ kg$

$W_A = mg$

$= 2(10)$

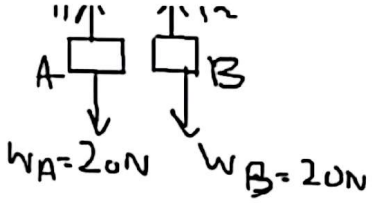
$= 20N$

$W_B = 20N$

A $T_1 = 20\ N$

B $T_2 = 20\ N$

At Pulley $T_3 = T_1 + T_2$



$W_A = mg$

$= 2(10)$

$= 20N$

$W_B = 20N$

A $T_1 = 20\ N$

B $T_2 = 20\ N$

At Pulley $T_3 = T_1 + T_2$

$= 20 + 20$

$= 40N$

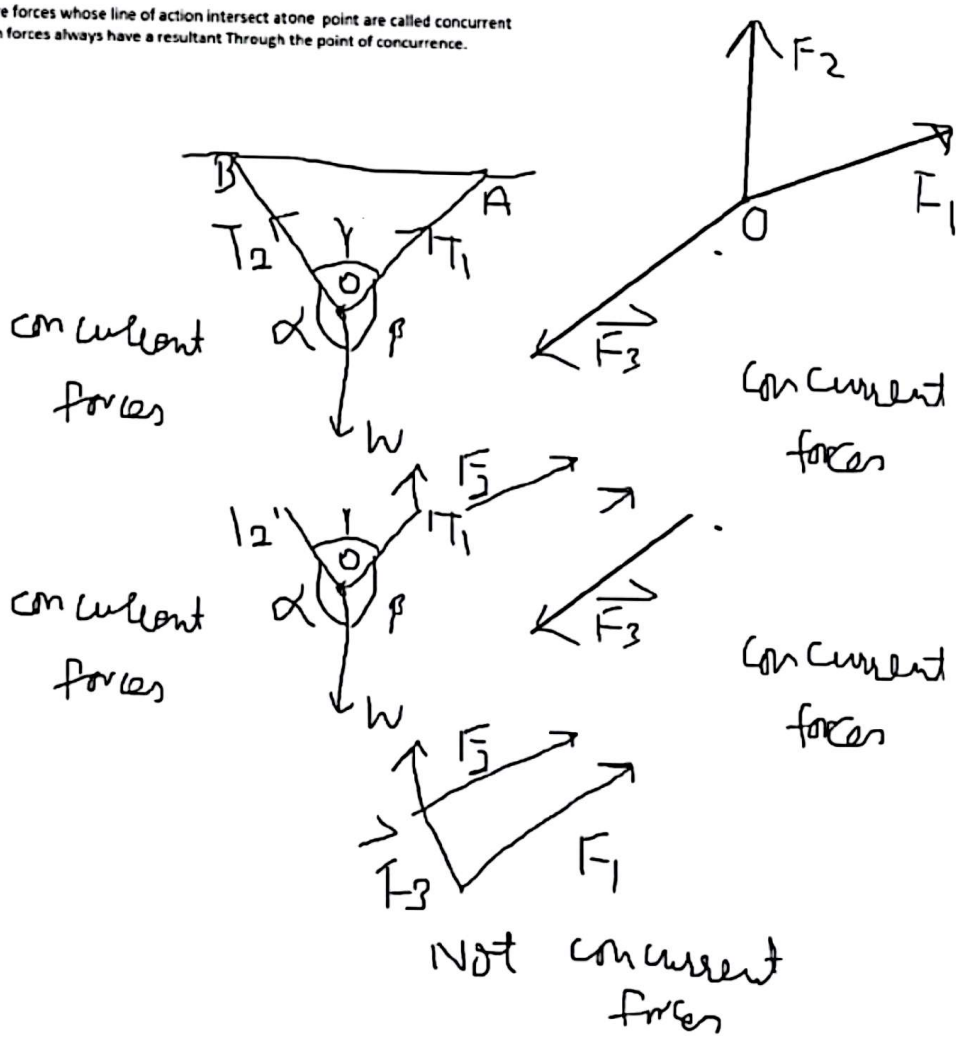
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Lecture No 09

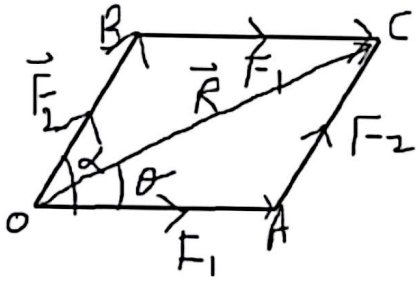
Concurrent Forces

Two or more forces whose line of action intersect atone point are called concurrent forces. Such forces always have a resultant Through the point of concurrence.



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Resultant of two forces acting at a point



$$\vec{R} = \vec{F_1} + \vec{F_2} \quad \text{---(1)}$$

$$\vec{R}.\vec{R} = \vec{R}.(\vec{F_1} + \vec{F_2})$$

$$R^2 = (\vec{F_1} + \vec{F_2})(\vec{F_1} + \vec{F_2})$$

$$= \vec{F_1}.\vec{F_1} + \vec{F_2}.\vec{F_1} + \vec{F_1}.\vec{F_2} + \vec{F_2}.\vec{F_2}$$

$$R^2 = (\vec{F_1} + \vec{F_2})(\vec{F_1} + \vec{F_2})$$

$$= \vec{F_1}.\vec{F_1} + \vec{F_2}.\vec{F_1} + \vec{F_1}.\vec{F_2} + \vec{F_2}.\vec{F_2}$$

$$= F_1 F_1 \cos 0 + F_2 F_1 \cos\alpha + F_1 F_2 \cos\alpha + F_2 F_2 \cos 0$$

$$R^2 = F_1^2 + 2F_1F_2\cos\alpha + F_2^2$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha} \quad \text{---(2)}$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha} \quad \text{---(2)}$$

$$\vec{F_1}.\vec{R} = \vec{F_1}.(\vec{F_1} + \vec{F_2})$$

$$F_1 R\cos\theta = F_1^2 + \vec{F_1}.\vec{F_2}$$

$$F_1 R\cos\theta = F_1^2 + F_1F_2\cos\alpha$$

$$R\cos\theta = F_1 + F_2\cos\alpha \quad \text{---(3)}$$

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$$= F_1 + F_2 \cos\alpha \quad \text{——— 3}$$

$$|\vec{F_1} \times \vec{R}| = |\vec{F_1} \times (\vec{F_1} + \vec{F_2})|$$

$$F_1 R \sin\theta = |\vec{F_1} \times \vec{F_1} + \vec{F_1} \times \vec{F_2}|$$

$$F_1 R \sin\theta = |\vec{F_1} \times \vec{F_2}|$$

$$F_1 R \sin\theta = F_1 \times F_2 \sin\alpha$$

$$F_1 R \sin\theta = |\vec{F_1} \times \vec{F_2}|$$

$$F_1 R \sin\theta = F_1 \times F_2 \sin\alpha$$

$$R \sin\theta = F_2 \sin\alpha \quad \text{——— (4)}$$

$$\frac{R\sin\theta}{R\cos\theta} = \frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha}$$

$$\frac{R\sin\theta}{R\cos\theta} = \frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha}$$

$$\tan\theta = \frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha}$$

$$\theta = \tan^{-1}\left(\frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha}\right)$$

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$$\theta = \tan^{-1}\left(\frac{F_2 \sin\alpha}{F_1 + F_2\cos\alpha}\right) \quad \text{---(5)}$$

Case-1

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha}$$

R is max $\cos\alpha = 1 \Rightarrow \alpha = 0^\circ$

$$\therefore R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2}$$

$$= \sqrt{(F_1 + F_2)^2}$$

$$R = F_1 + F_2$$

Case-2

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha}$$

$\cos\alpha = -1 \Rightarrow R_{min}$

$\Rightarrow \alpha = 180^\circ$

$$R_{min} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2(-1)}$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha}$$

$\cos\alpha = -1 \Rightarrow R_{min}$

$\Rightarrow \alpha = 180^\circ$

$$R_{min} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2(-1)}$$

$$= \sqrt{(F_1 - F_2)^2}$$

$$= |F_1 - F_2|$$

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$K_{min} = |F_1 - F_2|$

Case-3

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\alpha}$$

$\alpha = 90^\circ$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 90^\circ} \qquad \because \cos 90^\circ = 0$$

$$R = \sqrt{F_1^2 + F_2^2}$$

$\alpha = 90^\circ$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos 90^\circ} \qquad \because \cos 90^\circ = 0$$

$$R = \sqrt{F_1^2 + F_2^2}$$

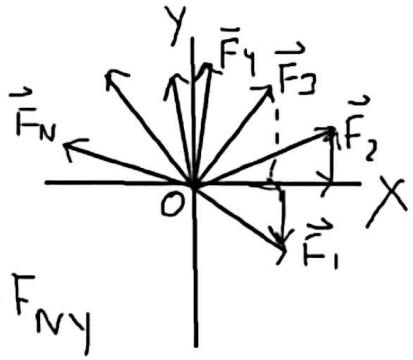
$$\theta = \tan^{-1}\left(\frac{F_2 \sin 90^\circ}{F_1 + F_2\cos 90^\circ}\right) \qquad \sin 90^\circ = 1$$

$$\theta = \tan^{-1}\left(\frac{F_2}{F_1}\right)$$

Lecture No 10

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Resultant of system of concurrent coplaner forces



$$R_x = F_{1x} + F_{2x} + F_{3x} + \cdots + F_{Nx}$$

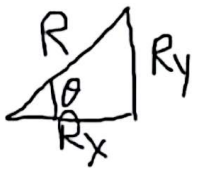
$$R_y = F_{1y} + F_{2y} + F_{3y} + \cdots + F_{Ny}$$

$$R^2 = R_x^2 + R_y^2$$

$$R_y = F_{1y} + F_{2y} + F_{3y} + \cdots + F_{Ny}$$

$$R^2 = R_x^2 + R_y^2$$

$$R = \sqrt{R_x^2 + R_y^2} \quad \text{---- (1)}$$

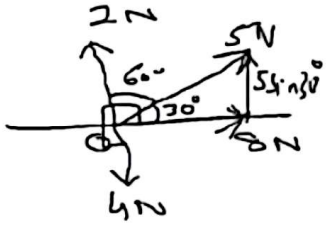


$$\tan\theta = \frac{R_y}{R_x}$$

$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right) \quad \text{---- (2)}$$

Example.

Find the magnitude and direction of the Resultant of the forces shown in the fig.



Solution

$$X = 8\cos 0^\circ + 5\cos 30^\circ + 2\cos 90^\circ + 4\cos 270^\circ$$

$$= 8(1) + 5(0.866) + 2(0) + 4(0)$$

$$= 8 + 4.33 + 0 + 0 = 12.33N$$

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$$Y = 8\sin 0^\circ + 5\sin 30^\circ + 2\sin 90^\circ + 4\sin 270^\circ$$

$$= 8(0) + 5\left(\frac{1}{2}\right) + 2(1) + 4(-1)$$

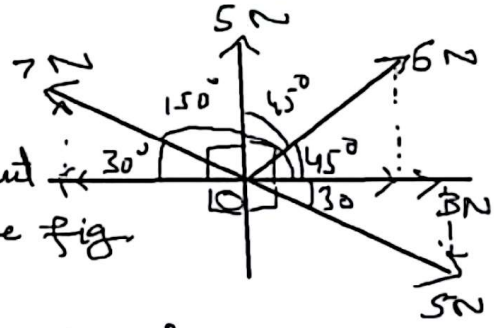
$$= 2.5 + 2 - 4 = -0.5\,N$$

$$R = \sqrt{x^2+y^2} = \sqrt{12.33^2 + 0.5^2} = \sqrt{152.27} = 12.34\,N$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{0.5}{12.33}\right) = 2.3^\circ \text{ with } 8N \text{ force.}$$

Example:

Find the magnitude and direction of the resultant of the forces shown in the fig.



Sol:

$$\rightarrow \quad X = 3\cos 0^\circ + 6\cos 45^\circ + 5\cos 90^\circ + 7\cos 150^\circ + 5\cos 330^\circ$$

$$= 3 + 4.243 + 0 - 6.062 + 4.330$$

$$= 5.511\,N$$

$$= 3 + 4.243 + 0 - 6.062 + 4.330$$

$$= 5.511\,N$$

$$\uparrow \quad Y = 3\sin 0^\circ + 6\sin 45^\circ + 5\sin 90^\circ + 7\sin 150^\circ + 5\sin 330^\circ$$

$$= 0 + 4.243 + 5 + 3.5 - 2.5$$

$$= 10.243\,N$$

$$R = \sqrt{X^2 + Y^2}$$

$$=$$

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$= 10.243\,N$

$$R = \sqrt{X^2 + Y^2}$$

$$= \sqrt{5.511^2 + 10.243^2}$$

$$= 11.63N$$

$$= \sqrt{5.511^2 + 10.243^2}$$

$$= 11.63N$$

$$\theta = \tan^{-1}\frac{Y}{X} = \tan^{-1}\left(\frac{10.243}{5.511}\right)$$

$= 61.7^0$ with 3N force

Lecture No 11

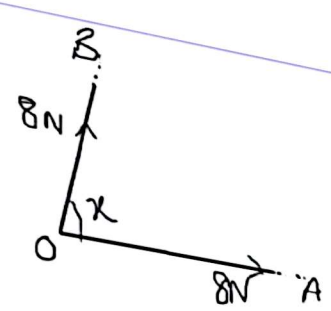
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Example

Two forces each of magnitude 8N act at a point in the direction OA and OB. The angle between the forces is x. The resultant of the two forces has component 9N in the direction OA. Find

(i) the value of x

(ii) The magnitude of resultant of the two forces.



Sol

$$R_x = 8N + 8\cos x$$

$$9N = 8N + 8\cos xN$$

$$8\cos x = 1$$

$$\cos x = \frac{1}{8}$$

$$8\cos x = 1$$

$$\cos x = \frac{1}{8}$$

$$x = 82.8^\circ$$

(ii)

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos x}$$

$$= \sqrt{8^2 + 8^2 + 2\times8\times8\times\cos 82.8}$$

(ii)

$$x = 82.8^\circ$$

$$R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos x}$$

$$= \sqrt{8^2 + 8^2 + 2\times8\times8\times\cos 82.8}$$

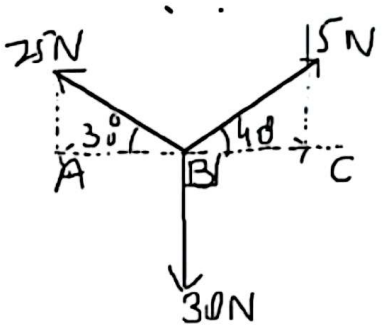
$$= \sqrt{64 + 64 + 128\cos 82.8}$$

$$= 12\ N$$

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Example

Coplaner forces of magnitudes 15N, 25N and 30N act at a point B on the line ABC in the direction shown in the diagram.
(i) find the magnitude and direction of the resultant force.
(ii)The force of magnitude 15N is now replaced by a force of magnitude FN acting in the same direction. The new resultant force has zero component in the direction BC. Find the value of F and also find the magnitude and direction of the new resultant force.



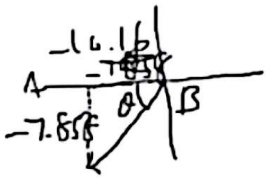
Sol:

$$R_x = 15\cos 40^\circ - 25\cos 30^\circ$$
$$= -10.16$$

$$R_y = 25\sin 30^\circ + 15\sin 40^\circ - 30$$

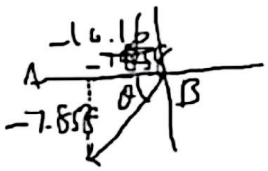
$$R_y = 25\sin 30^\circ + 15\sin 40^\circ - 30$$
$$= -7.858$$

$$R = \sqrt{R_x^2 + R_y^2}$$
$$= \sqrt{(-10.16)^2 + (-7.858)^2}$$
$$= 12.8\ N$$



$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)$$

$$R = \sqrt{R_x^2 + R_y^2}$$
$$= \sqrt{(-10.16)^2 + (-7.858)^2}$$
$$= 12.8\ N$$



$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)$$
$$= \tan^{-1}\left(\frac{7.858}{10.16}\right) = 37.7^\circ$$ with BA downward

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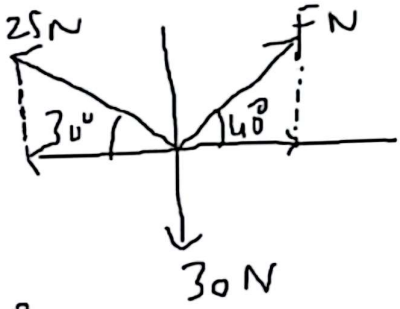
downward

(ii)

$F\cos 40^\circ = 25\cos 30^\circ$

$F = \frac{25\cos 30^\circ}{\cos 40^\circ}$

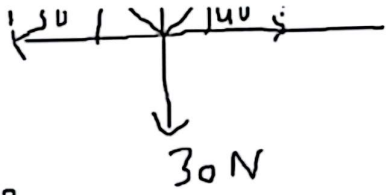
$= 28.3\ N$



$R_y = F\sin 40^\circ + 25\sin 30^\circ - 30$

$\frac{\quad\quad}{\cos 40^\circ}$

$= 28.3\ N$



$R_y = F\sin 40^\circ + 25\sin 30^\circ - 30$

$= 28.3\sin 40^\circ + 25\sin 30^\circ - 30$

$= 0.667\ N$ upward

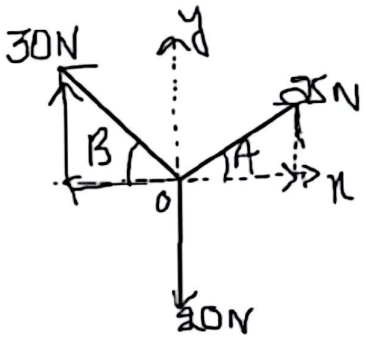
Example

Three coplaner forces act at a point. The magnitude of the forces are 20N, 25N and 30N and the direction in which forces act are shown in the diagram, where sinA=0.28 and cosA=0.96 and sinB=0.6 and cosB=0.8

(i) Show that the resultant of the three forces has a zero-component in the x-direction.

(ii) Find the magnitude and direction of the resultant of the three forces.

(iii) The force of magnitude 20N is replaced by another force. The effect is that the resultant force is unchanged in the magnitude but reverse in direction. State the magnitude and the direction of the replacement force



Sol

$R_x = \sum F_x$

$= 25\cos A - 30\cos B$

$= 25(0.96) - 30(0.8)$

$= 24 - 24 = 0$

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$= 25(0.96) - 30(0.8)$
$= 24 - 24 = 0$

(ii) $R_y = \Sigma F_y$
$= 25 \sin A + 30 \sin B - 20$
$= 25(0.28) + 30(0.6) - 20$
$= 5\ N$

$= 25(0.28) + 30(0.6) - 20$
$= 5\ N$

$R = \sqrt{R_x^2 + R_y^2}$
$= \sqrt{0^2 + (5N)^2}$
$= \sqrt{25}$
$= 5\ N$

$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)$
$= \tan^{-1}\left(\frac{5}{0}\right)$
$= 90^\circ$

$K = \sqrt{K_x^2 + R_y^2}$
$= \sqrt{0^2 + (5N)^2}$
$= \sqrt{25}$
$= 5\ N$

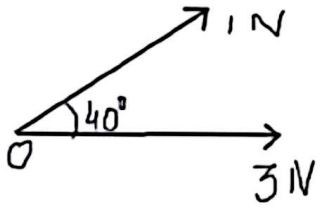
$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)$
$= \tan^{-1}\left(\frac{5}{0}\right)$
$= 90^\circ$

(iii) $X = 30\ N$ downward

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Lecture No 12

Example
Two forces each of magnitude 1N and 3N act on a particle in the direction as shown in the diagram. Calculate the resultant force on the particle and the angle between the resultant and 3N force.



Sol:-

$$R = \sqrt{P^2 + q^2 + 2pq\cos\theta}$$

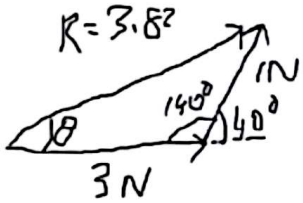
$$= \sqrt{1^2 + 3^2 + 2(1)(3)\cos 40^\circ}$$

$$= \sqrt{1 + 9 + 6(0.7660)}$$

$$= \sqrt{1 + 9 + 6(0.7660)}$$

$$= \sqrt{14.5983}$$

$$= 3.82\ N$$

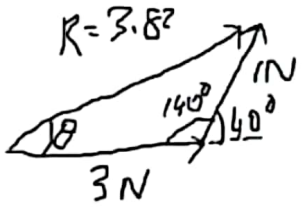


Using Sine rule

$$\frac{\sin\theta}{1} = \frac{\sin 140^\circ}{3.82}$$

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$= 3.82\ N$



Using Sine rule

$$\frac{\sin\theta}{1} = \frac{\sin 140^\circ}{3.82}$$

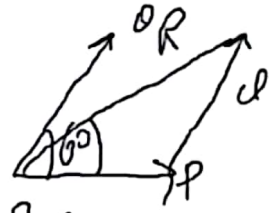
$$\sin\theta = 0.1681$$

$$\theta = \sin^{-1}[0.1682]$$

$$= 9.68^\circ = 9.7^\circ \text{ with 3N Force}$$

Example

Find the magnitude of two forces such that if they act at right angles, their resultant is $\sqrt{10}$ N and when the act at an angle of 60 degrees, their resultant is $\sqrt{13}$ N.

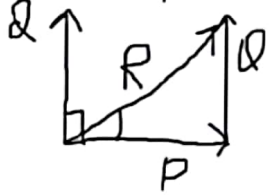


Sol:- Let the force be P and q

$$R^2 = P^2 + q^2$$

$$(\sqrt{10})^2 = P^2 + q^2$$

$$P^2 + q^2 = 10 \quad \text{——(1)}$$



$$P^2 + q^2 = 10 \quad \text{——(1)}$$



$$R^2 = P^2 + q^2 + 2Pq\cos\theta$$

$$(\sqrt{13})^2 = P^2 + q^2 + 2Pq\cos 60^\circ$$

$$13 = 10 + 2Pq\left(\frac{1}{2}\right)$$

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$$R^2 = P^2 + Q^2 + 2PQ\cos\theta$$

$$(\sqrt{13})^2 = P^2 + Q^2 + 2PQ\cos 60$$

$$13 = 10 + 2PQ\left(\frac{1}{2}\right)$$

$$3 = PQ$$

$$PQ = 3 \quad \text{---(2)}$$

$$Q = \frac{3}{P} \quad \text{---(3)}$$

$$Q = \frac{3}{P} \quad \text{---(3)}$$

(1) =>

$$P^2 + \left(\frac{3}{P}\right)^2 = 10$$

$$P^2 + \frac{9}{P^2} = 10$$

$$P^4 + 9 = 10P^2$$

$$P^4 - 10P^2 + 9 = 0$$

$$P^2 = t \quad \text{---(4)}$$

$$t^2 - 10t + 9 = 0$$

$$t^2 - 9t - t + 9 = 0$$

$$t(t-9) - 1(t-9) = 0$$

$$(t-9)(t-1) = 0$$

$$t = 9, \; t = 1$$

$P^4 + 9 = 10P^2$

$P^4 - 10P^2 + 9 = 0$

$P^2 = t \quad \text{---(4)}$

$t^2 - 10t + 9 = 0$

$t = 1 \Rightarrow P^2 = 1 \Rightarrow P = 1$

$t = 9 \Rightarrow P^2 = 9 \Rightarrow P = 3$

$P = 1 \qquad P = 3$

using (3)

$Q = \frac{3}{1} = 3$

$Q = \frac{3}{3} = 1$

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$p^2 = t$ ——(4)

$t^2 - 10t + 9 = 0$

$t^2 - 9t - t + 9 = 0$

$t(t-9) - 1(t-9) = 0$

$(t-9)(t-1) = 0$

$t = 9, \; t = 1$

$t = 9 \Rightarrow p^2 = 9 \Rightarrow p = 3$

$p = 1$

using (3)

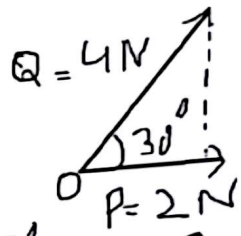
$q = \frac{3}{1} = 3$

$p = 3$

$q = \frac{3}{3} = 1$

3N, 1N

Example
Two forces each of magnitude 2N and 4N act on a point on the particle in the direction as shown in the diagram. Calculate the angle between the resultant and 2N force.

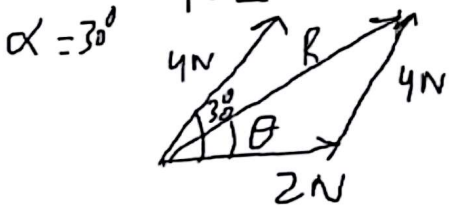


Sol:

$$\tan\theta = \frac{Q\sin\alpha}{P + Q\cos\alpha}$$

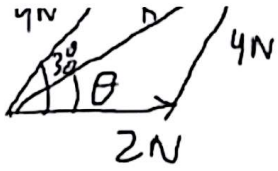
$$\tan\theta = \frac{4\sin 30^0}{2 + 4\cos 30^0}$$

$$=$$



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$$\tan\theta = \frac{P + Q\cos\alpha}{4\sin 30^\circ}$$

$$= \frac{2 + 4\cos 30^\circ}{4\left(\frac{1}{2}\right)}$$

$$= \frac{2 + 4(0.866)}{2}$$

$$= \frac{5.46}{2}$$

$$= 2.73$$

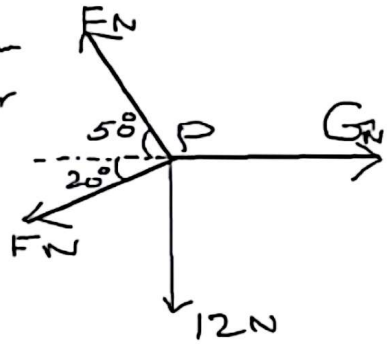
$$\theta = \tan^{-1}(2.73)$$

$$= 70.1^\circ$$

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Lecture No 13

Example:- A particle P is in equilibrium on a smooth horizontal table under the action of horizontal forces of the magnitude FN, FN and 12N acting in the direction shown. Find the value of F and G



Sol:-

$$X = 0 \quad \text{and} \quad Y = 0$$

the action of horizontal forces of the magnitude FN, FN and 12N acting in the direction shown. Find the value of F and G



Sol:-

$$X = 0 \quad \text{and} \quad Y = 0$$

$$Y = 0$$

$$F\sin 50^\circ = F\sin 20^\circ + 12$$

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$X=0$ and $Y=0$

$Y=0$

$F\sin 50^\circ = F\sin 20^\circ + 12$

$F\sin 50^\circ - F\sin 20^\circ = 12$

$F(\sin 50^\circ - \sin 20^\circ) = 12$

$F = \dfrac{12}{\sin 50^\circ - \sin 20^\circ}$

$F = 28.3\,N$

$X = 0$

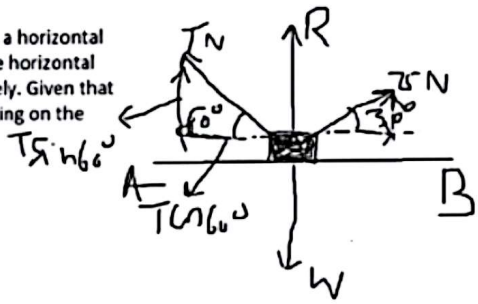
$G = F\cos 50^\circ + F\cos 20^\circ$

$= F(\cos 50^\circ + 20^\circ)$

$= 28.3(\cos 50^\circ + \cos 20^\circ)$

$= 44.78$

Ex.
Two Light strings are attached to a block of mass 20kg. The block is in equilibrium on a horizontal Surface AB with its strings taut. The strings make angle of 60 and 30 degrees with the horizontal on either sides of the block and the tensions in the strings are TN and 75N respectively. Given that the surface is smooth, find the value of T and the magnitude of the contact force acting on the block.



Solution

$m = 20\,kg$

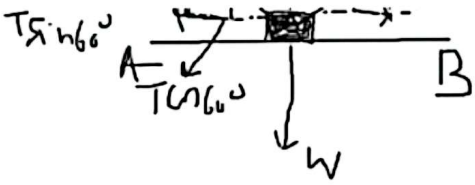
$W = mg$

$= 20(10)$

$= 200\,N$

$X = 0$

block.



Solution

$m = 20\,kg$

$W = mg$

$= 20(10)$

$= 200\,N$

$X = 0$

$\Rightarrow T\cos 60^\circ = 75\cos 30^\circ$

$T = \dfrac{75\cos 30^\circ}{\cos 60^\circ}$

$= 129.9\,N$

$Y = 0$

$R + T\sin 60^\circ + 75\sin 30^\circ = 200$

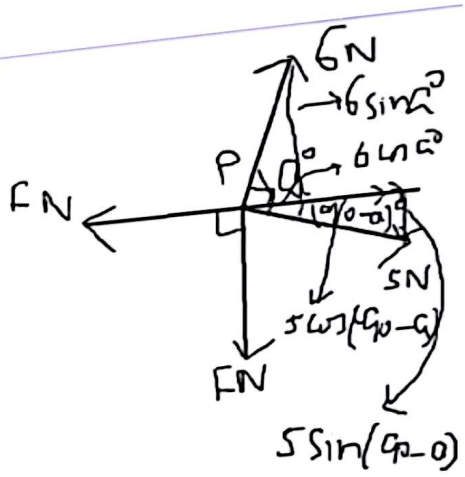
$R + 129.9\sin 60^\circ + 75\sin 30^\circ = 200$

$R = 200 - 129.9\sin 60^\circ - 75\sin 30^\circ$

$R = 50\,N$

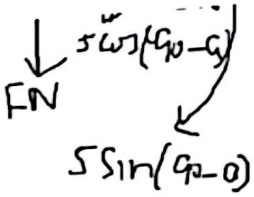
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Example
A Particle P is in equilibrium on a smooth horizontal table under the action of horizontal forces of magnitude 6N, 5N, FN and FN acting in the direction shown. Find the value of a and F.



Sol

$$F = 6\cos a + 5\cos(90-a)$$
$$= 6\cos a + 5\sin a \quad \text{---(1)}$$



$$= 6\cos a + 5\sin a \quad \text{---(1)}$$

$$F + 5\sin(90-a) = 6\sin a^\circ$$
$$F + 5\cos a = 6\sin a^\circ$$
$$F = 6\sin a - 5\cos a \quad \text{---(2)}$$
$$6\sin a - 5\cos a = 6\cos a + 5\sin a$$

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$$F + 5\sin(90-\alpha) = 6\sin\alpha^\circ$$

$$F + 5\cos\alpha = 6\sin\alpha^\circ$$

$$F = 6\sin\alpha - 5\cos\alpha \quad \text{---(2)}$$

$$6\sin\alpha - 5\cos\alpha = 6\cos\alpha + 5\sin\alpha$$

$$6\sin\alpha - 5\sin\alpha = 6\cos\alpha + 5\cos\alpha$$

$$\sin\alpha = 11\cos\alpha = \frac{\sin\alpha}{\cos\alpha} = 11$$

$$\tan\alpha = 11$$

$$\alpha = \tan^{-1}(11)$$

$$= 84.8^\circ$$

Using (1)

$$F = 6\cos 84.8^\circ + 5\sin 84.8$$

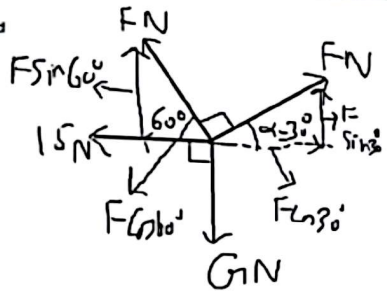
$$= 5.52N$$

Example

Four horizontal forces act a point O and are in equilibrium. the magnitude of the forces are F N, G N 15N, and F N. and the forces act in the direction as shown in the diagram.

(i) Show that F=41.0 correct to 3 significant figures.

(ii) Find the value of G.



Solution

$$\alpha + 60 + 90^\circ = 180^\circ$$

$$\alpha = 30^\circ$$

$$\rightarrow \quad F\cos 30^\circ = F\cos 60^\circ + 15$$

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$\alpha = 30°$

$F\cos 30° = T\cos 60° + 15$

$F\cos 30° - T\cos 60° = 15$

$F(\cos 30° - \cos 60°) = 15$

$F = \dfrac{15}{\cos 30° - \cos 60°}$

$F = 40.983$

$F(\cos 30° - \cos 60°) = 15$

$F = \dfrac{15}{\cos 30° - \cos 60°}$

$F = 40.983$

$F = 41.0$ correct to 3 s.f.

(ii)

$G = F\sin 30° + F\sin 60°$

$= 41\sin 30° + 41\sin 60°$ correct to 4 s.f.

(ii)

$G = F\sin 30° + F\sin 60°$

$= 41\sin 30° + 41\sin 60°$

$= 55.983$

$= 56.0\,N$

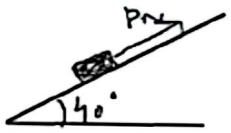
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Example
a small block of weight 12 N is at rest on a smooth plane inclined at 40 degrees to the horizonatal. The block is held in equilibrium by a force of magnitude PN. find the value of P when
(i) the force is parallel to the plane as in fig 1
(ii) The force is horizontal to the plane as in Fig 2.

Fig 1



Sol

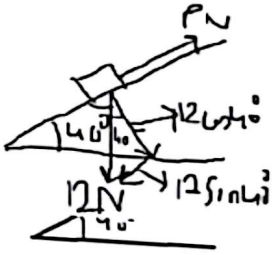
$P = 12 \sin 40^\circ$

$= 7.71 N$

Fig 2



Fig(i)

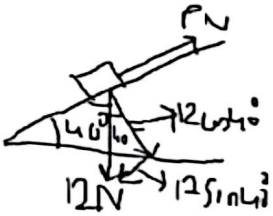


(ii) $P = 12 \sin 40^\circ$

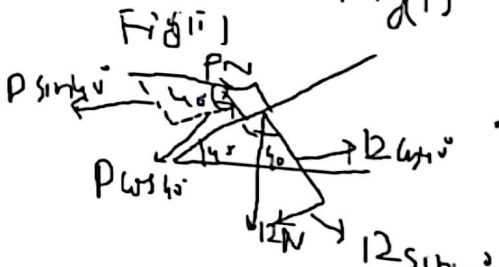
$= 7.71 N$

(ii)

Fig(i)



Fig(ii)



$P \cos 40^\circ = 12 \sin 40^\circ$

$P = \frac{12 \sin 40^\circ}{\cos 40^\circ}$

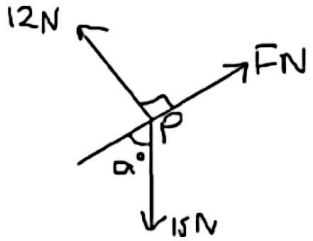
$P = 10.1 N$

Lecture No 15

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Example
Three coplaner forces of magnitudes F N, 12N and 15N are in equilibrium acting at a point P in the direction shown in the diagram. find a and F.



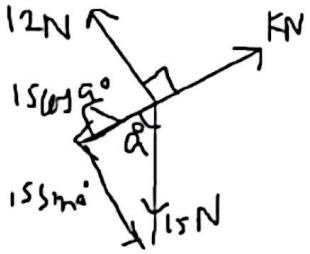
Sol:-

↑ $15\sin a = 12$

$\sin a = \frac{12}{15}$

$a = \sin^{-1}\left(\frac{12}{15}\right)$

$= 53.1^{\circ}$



Sol:-

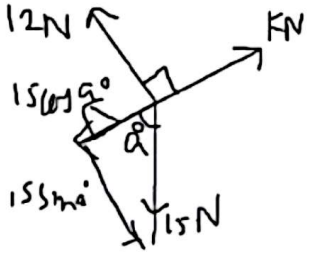


↑ $15\sin a = 12$

$\sin a = \frac{12}{15}$

$a = \sin^{-1}\left(\frac{12}{15}\right)$

$= 53.1^{\circ}$

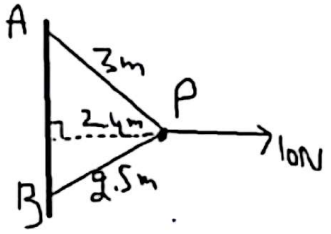


→ $T = 15\cos a^{\circ}$

$= 15 \times \cos 53.1^{\circ}$

$= 9N$

Example
A and B are fixed points of a vertical wall with A vertically above B. A particle P of mass 0.7 kg is attached to A by a light inextensible string of length 3m. P is also attached to B by a light inextensible string of length 2.5m. P is maintained in equilibrium at a distance of 2.4m from the wall by a horizontal force of magnitude 10N acting at P. Both strings are taut and the 10N force acts in the plane APB which is perpendicular to the wall. Find the tensions in the strings.

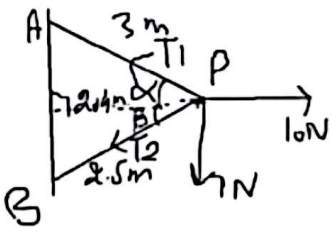


Sol:-

$m = 0.7\ kg$

$W = mg$

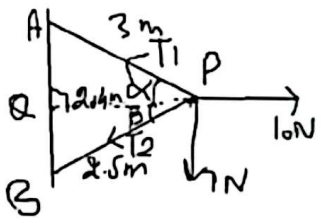
$= 0.7 \times 10 = 7N$



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$W = mg$ ∴ d

$= 0.7 \times 10 = 7N$



ΔAQB

$AQ^2 = AP^2 - QP^2$

$AQ = \sqrt{3^2 - 2.4^2} = 1.8\,m$

$\cos\alpha = \frac{2.4}{3} = \frac{24}{30} = \frac{4}{5}$, $\sin\alpha = \frac{1.8}{3} = \frac{18}{30} = \frac{3}{5}$

ΔQPB

$QB^2 = PB^2 - QP^2$

$QB = \sqrt{2.5^2 - 2.4^2} = 0.7$

$XD = PB - QP$

$QB = \sqrt{2.5^2 - 2.4^2} = 0.7$

$\cos\beta = \frac{2.4}{2.5} = 24/25$

$\cos\beta = \frac{2.4}{2.5} = 24/25$

$\sin\beta = \frac{0.7}{2.5} = \frac{7}{25}$

$\rightarrow$ $T_1\cos\alpha + T_2\cos\beta = 10N$

$T_1 \times \frac{4}{5} + T_2 \times \frac{24}{25} = 10$

$T_1 = \frac{5}{4}\left(10 - \frac{24}{25}T_2\right)$

$\cdot 1 \times \frac{1}{5} + {}'2 \times \frac{24}{25} = 10$

$T_1 = \frac{5}{4}\left(10 - \frac{24}{25}T_2\right)$ ——— (1)

$\downarrow$ $T_1\sin\alpha = T_2\sin\beta + 7$

$T_1 \times \frac{3}{5} = T_2 \times \frac{7}{25} + 7$

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$$T_1 = \frac{5}{3}\left(\frac{7T_2}{25} + 7\right) \quad \text{——(2)}$$

Using (1) & (2)

$$\frac{7}{4}\left(10 - \frac{24}{25}T_2\right) = \frac{8}{3}\left(\frac{7T_2}{25} + 7\right)$$

$$3\left(10 - \frac{24}{25}T_2\right) = 4\left(\frac{7T_2}{25} + 7\right)$$

$$3\left(10 - \frac{24}{25}T_2\right) = 4\left(\frac{7T_2}{25} + 7\right)$$

$$0.8T_2 = 0.4$$

$$T_2 = 0.5\,N$$

$$T_1 = \frac{5}{3}\left(\frac{7(0.5)}{25} + 7\right)$$

$$= 11.9\,N$$

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Lecture No 16

Lamy's theorem

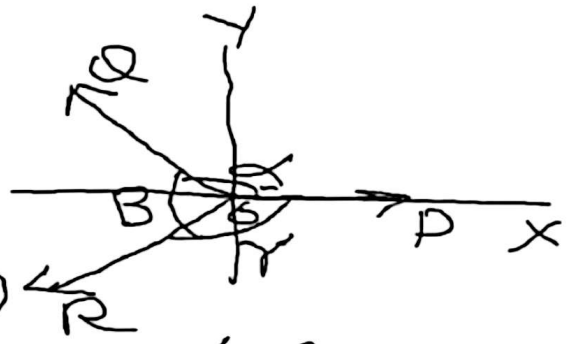
If a particle is in equilibrium under the action of three concurrent forces, then the magnitude of each force is proportional to the Sine of angle between other two.

Proof ..



... of each force is proportional to the Sine of angle between other two.

Proof ..



$$X = P\cos 0^\circ + Q\cos\alpha + R\cos(\alpha+\beta)$$

$$X = P(1) + Q\cos\alpha + R\cos\gamma$$

$$X = P + Q\cos\alpha + R\cos\gamma$$

$$\alpha+\beta+\gamma = 2\pi$$

$$\alpha+\beta = 2\pi-\gamma$$

$$\cos(\alpha+\beta) = \cos(2\pi-\gamma)$$

$$= \cos\gamma$$

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$X = P(1) + Q\cos\alpha + R\cos\gamma$

$x = P + Q\cos\alpha + R\cos\gamma$

$\alpha+\beta+\gamma = 2\pi$

$\alpha+\beta = 2\pi - \gamma$

$\cos(\alpha+\beta) = \cos(2\pi-\gamma)$

$= \cos\gamma$

$Y = P\sin 0^\circ + Q\sin\alpha + R\sin(\alpha+\beta)$

$= P(0) + Q\sin\alpha + R\sin(\alpha+\beta)$

$\alpha+\beta+\gamma = 2\pi$

$\alpha+\beta = 2\pi-\gamma$

$\sin(\alpha+\beta) = \sin(2\pi-\gamma)$

$Y = Q\sin\alpha - R\sin\gamma$ $= -\sin\gamma$

$\dot{X}=0$, $\dot{Y}=0$

$X=0$

$\Rightarrow P+Q\cos\alpha+R\cos\gamma=0$ ——(1)

$Y=0$

$\Rightarrow Q\sin\alpha - R\sin\gamma = 0$

$Q\sin\alpha = R\sin\gamma$

$\frac{R}{\sin\alpha} = \frac{Q}{\sin\gamma}$ ——(2)

(2) $\Rightarrow$ $Q = R\sin\gamma$

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$$\frac{R}{\sin\alpha} = \frac{Q}{\sin r} \quad \text{---(2)}$$

(2) $\Rightarrow$ $Q = \frac{R\sin r}{\sin\alpha}$ ---(3)

using (3) in (1)

$$P + R\frac{\sin r}{\sin\alpha}\cos\alpha + R\cos r = 0$$

$$P + R\left(\frac{\sin r}{\sin\alpha}\cos\alpha + \cos r\right) = 0$$

$$P + R\left(\frac{\sin r\cos\alpha + \cos r\sin\alpha}{\sin\alpha}\right) = 0$$

$$P + R\frac{\sin r}{\sin\alpha}\cos\alpha + R\cos r = 0$$

$$P + R\left(\frac{\sin r}{\sin\alpha}\cos\alpha + \cos r\right) = 0$$

$$P + R\left(\frac{\sin r\cos\alpha + \cos r\sin\alpha}{\sin\alpha}\right) = 0$$

$$P + R\frac{\sin(\alpha + r)}{\sin\alpha} = 0$$

$$P + R\frac{(-\sin\beta)}{\sin\alpha} = 0$$

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$$T + R\frac{\sin(\alpha+\gamma)}{\sin\alpha} = 0$$

$$P + R\frac{(-\sin\beta)}{\sin\alpha} = 0$$

$$P = \frac{R\sin\beta}{\sin\alpha}$$

$$\frac{P}{\sin\beta} = \frac{R}{\sin\alpha} \quad \text{——— (4)}$$

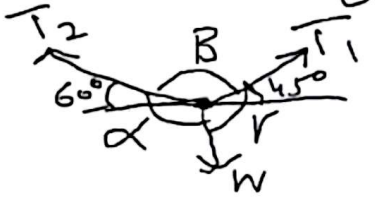
Comb. eq. (2) & (4)

$$\frac{P}{\sin\beta} = \frac{Q}{\sin\gamma} = \frac{R}{\sin\alpha}$$

Example A particle of mass 15 Kg is suspended by two strings. If the angles made by string are 45° and 60° with the horizontal. Find the Tensions in the string.

Sol

$$60^\circ + \beta + 45^\circ = 180^\circ$$
$$\beta = 180^\circ - 60^\circ - 45^\circ$$
$$= 75^\circ$$



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$\beta = 180^\circ - 60^\circ - 45^\circ$
$= 75^\circ$

$\alpha = 60^\circ + 90^\circ$
$= 150^\circ$

$\gamma = 90^\circ + 45^\circ$
$\gamma = 135^\circ$

$W = mg$
$= 15 \times 10$
$= 150N$

$$\frac{T_1}{\sin\alpha} = \frac{T_2}{\sin\gamma} = \frac{W}{\sin\beta}$$

$$\frac{T_1}{\sin\alpha} = \frac{T_2}{\sin\gamma} = \frac{W}{\sin\beta}$$

$= 15 \times 10$
$= 150N$

$$\frac{T_1}{\sin\alpha} = \frac{W}{\sin\beta}$$

$$\frac{T_1}{\sin 150^\circ} = \frac{150}{\sin 75^\circ}$$

$$T_1 = \frac{150 \sin 150^\circ}{\sin 75^\circ} = \frac{150 \times 0.5}{0.9659} = 77.7N$$

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$$T_1 = \frac{120 \sin 150}{\sin 75^\circ} = \frac{150 \times 0.5}{0.9659} = 77.7\,N$$

$$\frac{T_2}{\sin \gamma} = \frac{W}{\sin \beta}$$

$$\frac{T_2}{\sin 135^\circ} = \frac{150}{\sin 75^\circ}$$

$$T_2 = \frac{150 \times \sin 135^\circ}{\sin 75^\circ}$$

$$T_2 = \frac{150 \times \sin 135^\circ}{\sin 75^\circ}$$

$$= \frac{150 \times 0.707}{0.9659}$$

$$= 109.8\,N$$

$T_1 = 77.7N$

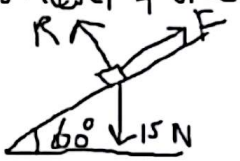
$T_2 = 109.8\,N$

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Lecture No 17

Example A block of mass 1.5Kg rests in Equilibrium in a rough inclined plane at 60° to the horizontal. Find the magnitude of the frictional contact force and normal contact force



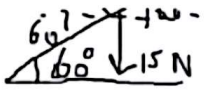
Sol:- $m = 1.5Kg$

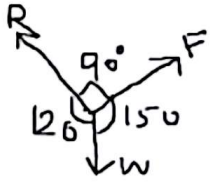
$W = mg$

$= 1.5 \times 10 = 15N$

$W = mg$

$= 1.5 \times 10 = 15N$





$$\frac{F}{\sin 120^\circ} = \frac{R}{\sin 150^\circ} = \frac{15}{\sin 90^\circ}$$

$$\frac{F}{\sin 120^\circ} = \frac{15}{\sin 90^\circ}$$

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$$\frac{T}{\sin 120^\circ} = \frac{15}{\sin 90^\circ}$$

$$T = \frac{15}{1} \times \sin 120^\circ$$

$$= 15 \times 0.866$$

$$= 13N$$

$$\frac{R}{\sin 150} = \frac{15}{\sin 90^\circ}$$

$$= 13N$$

$$\frac{R}{\sin 150} = \frac{15}{\sin 90^\circ}$$

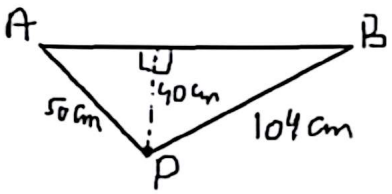
$$R = \frac{15 \times \sin 30^\circ}{\sin 90^\circ}$$

$$= \frac{15 \times 0.5}{1}$$

$$= 7.5N$$

Example

A particle P of mass 2.1kg is attached to one end of each of two light inextensible strings. The other ends of the string are attached to points A and B which are at some horizontal level. P hangs in equilibrium at a point 40m below the level of A and B, and the strings PA and PB have lengths 50cm and 104cm respectively as shown in the diagram. Show that the tension in the string PA is 20N, and find the tension in the string PB.



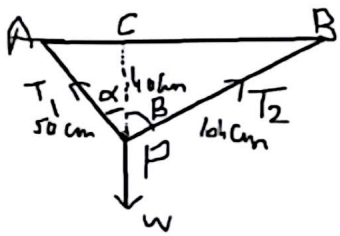
Sol

T_1 = Tension in PA

T_2 = Tension in PB

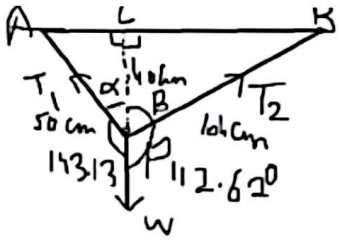
$m = 2.1kg$

$W = mg = 2.1(10) = 21N$



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T_1 = Tension in PA
T_2 = Tension in PB
$m = 2.1\ kg$
$W = mg = 2.1(10) = 21N$



ΔACP $\cos\alpha = \frac{PC}{PA} = \frac{40}{50} \Rightarrow \alpha = 36.87°$

ΔBCP $\cos B = \frac{PC}{PB} = \frac{40}{104} \Rightarrow B = 67.38°$

$\alpha + B = 36.87° + 67.38° = 104.25°$

$PB = 104 \Rightarrow B = 61.56$

$\alpha + B = 36.87° + 67.38° = 104.25°$

By Lami Th.

$$\frac{T_1}{\sin 112.62°} = \frac{T_2}{\sin 143.13°} = \frac{W}{\sin 104.25°}$$

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By Lami Th.

$$\frac{T_1}{\sin 112.62^\circ} = \frac{T_2}{\sin 143.13^\circ} = \frac{W}{\sin 104.25^\circ}$$

$$\frac{T_1}{\sin 112.62^\circ} = \frac{21}{\sin 104.25^\circ} \Rightarrow T_1 = \frac{21 \times \sin 112.62^\circ}{\sin 104.25^\circ}$$

$$T_1 = 20N$$

$$\frac{T_1}{\sin 112.62^\circ} = \frac{21}{\sin 104.25^\circ} \Rightarrow T_1 = \frac{21 \times \sin 112.62^\circ}{\sin 104.25^\circ}$$

$$T_1 = 20N$$

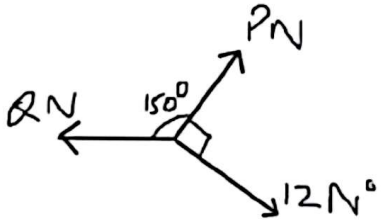
$$\frac{T_2}{\sin 143.13^\circ} = \frac{21}{\sin 104.25^\circ} \Rightarrow T_2 = \frac{21}{\sin 104.25^\circ} \times \sin 143.13^\circ$$

$$T_2 = 13N$$

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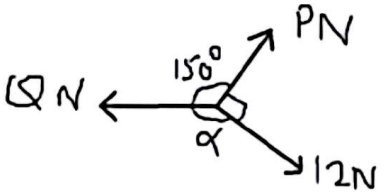
Lecture No 18

Example
Three coplanarforcesP N, Q N and 12 N are acting on a particle and are in equilibrium as shown in the figure. Find the magniutdes of the forces P and Q.

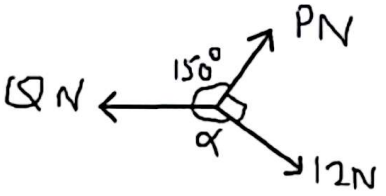


Sol:-

$$\alpha + 90^\circ + 150^\circ = 360^\circ$$
$$\alpha + 240^\circ = 360$$
$$\alpha = 360 - 240$$
$$= 120^\circ$$



$$\alpha = 360 - 240$$
$$= 120^\circ$$



$$\frac{P}{\sin 120^\circ} = \frac{Q}{\sin 90^\circ} = \frac{12}{\sin 150^\circ}$$

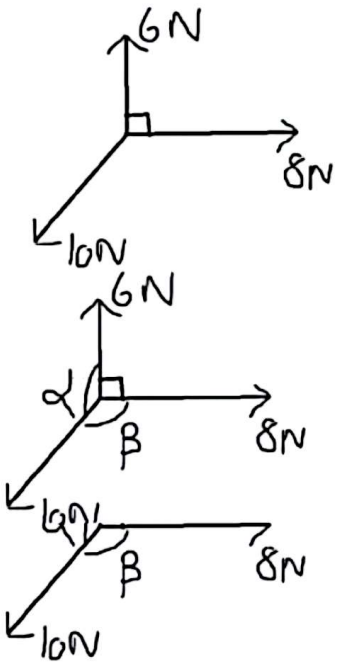
$$\frac{P}{\sin 120^\circ} = \frac{12}{\sin 150^\circ} \Rightarrow P = \frac{\sin 120^\circ \times 12}{\sin 150^\circ} \Rightarrow P = 20.8\ N$$

$$\frac{Q}{\sin 90^\circ} = \frac{12}{\sin 150^\circ} \Rightarrow Q = \frac{12 \sin 90^\circ}{\sin 150^\circ} \Rightarrow Q = 24\ N$$

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Example

Three coplanar forces are acting at a point O as shown in the fig. The magniutedes of the forces are 6N, 8N and 10N respectively. The angle between the 6N force and the 8N force is 90 degrees. The forces are in equilibrium. Find the other angles between the forces.



Sol

$$\frac{\sin\alpha}{8} = \frac{\sin B}{6} = \frac{\sin 90^\circ}{10}$$

$$\frac{\sin\alpha}{8} = \frac{\sin 90^\circ}{10} \Rightarrow \sin\alpha = \frac{8}{10} = 0.8$$

$$\alpha = 180 - 53.13^\circ = 126.87^\circ$$

$$B + \alpha + 90^\circ = 360^\circ$$

$$B + 126.87^\circ + 90^\circ = 360^\circ$$

$$B = 360^\circ - 90^\circ - 126.87^\circ$$

$$= 270 - 126\ 87$$

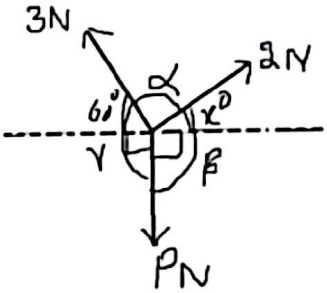
$$B = 143.13^\circ$$

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Lecture No 19

Example
The three concurrent coplanar forces shown in the diagram have magnitudes 3N, 2N and PN. Given that the three forces are in equilibrium, find the value of angle x and P.



Sol

$\gamma = 60^\circ + 90^\circ = 150^\circ$

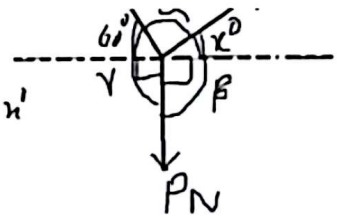
$\frac{P}{\sin\alpha} = \frac{3}{\sin\beta} = \frac{2}{\sin 150^\circ}$

So $\frac{3}{\sin\beta} = \frac{2}{\sin 150}$

Sol

$\gamma = 60^\circ + 90^\circ = 150^\circ$

$\frac{P}{\sin\alpha} = \frac{3}{\sin\beta} = \frac{2}{\sin 150^\circ}$



$\beta = 90^\circ + x^\circ$

So $\frac{3}{\sin\beta} = \frac{2}{\sin 150} \Rightarrow 3 \times \sin 150^\circ = 2\sin\beta$

$\sin\beta = \frac{3}{2\times 2} = \frac{3}{4} \Rightarrow \beta = \sin^{-1}\left(\frac{3}{4}\right) = 48.6$

$\because \beta$ is obtuse $\therefore \beta = 180 - 48.6 = 131.4^\circ$

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β is obtuse $\therefore \beta = 180 - 48.6 = 131.4^\circ$

$\gamma = \beta - 90^\circ = 131.4^\circ - 90^\circ = 41.4^\circ \Rightarrow x = 41.4^\circ$

$\alpha + \beta + \gamma = 360^\circ$

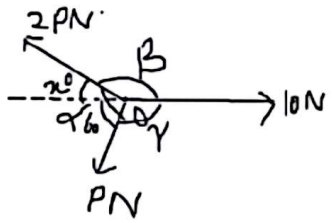
$\alpha + 131.4^\circ + 150^\circ = 360^\circ \Rightarrow \alpha = 78.6^\circ$

$$\frac{P}{\sin\alpha} = \frac{2}{\sin 150^\circ} \Rightarrow \frac{P}{\sin 78.6^\circ} = \frac{2}{\sin 150^\circ}$$

$P = 4 \times \sin 78.6^\circ$

$P = 3.92\,N$

Example
The three coplanar forces shown in the diagram are in equilibrium. Find the values of angle x and force P.



Sol

$\gamma + 60^\circ = 180^\circ \qquad \alpha = x^\circ + 60^\circ$

$\gamma = 120^\circ$

$$\frac{P}{\sin\beta} = \frac{2P}{\sin 120^\circ} = \frac{10}{\sin\alpha}$$

$$\frac{P}{\sin\beta} = \frac{2P}{\sin 120^\circ}$$

$\sin\beta = \sin 120^\circ = \frac{\sqrt{3}}{2} \Rightarrow \beta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \Rightarrow \beta = 25^\circ 7'$

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$$\frac{1}{\sin\beta} = \frac{2P}{\sin 120^\circ} = \frac{10}{\sin\alpha} \quad \ldots$$

$$\frac{P}{\sin\beta} = \frac{2P}{\sin 120^\circ}$$

$$\sin\beta = \frac{\sin 120^\circ}{2} = \frac{\sqrt{3}}{4} \Rightarrow \beta = \sin^{-1}\left(\frac{\sqrt{3}}{4}\right) \Rightarrow \beta = 25.7^\circ$$

$$\beta = 180 - 25.7^\circ = 154.3^\circ$$

$$\gamma = 180 - \beta = 180 - 154.3^\circ \Rightarrow \gamma = 25.7^\circ$$

$$\alpha = x^\circ + 60^\circ = 60 + 25.7^\circ = 85.7^\circ$$

$$\alpha = x^\circ + 60^\circ = 60 + 25.7^\circ = 85.7^\circ$$

$$\frac{2P}{\sin 120^\circ} = \frac{10}{\sin\alpha}$$

$$2P = \frac{10 \times \sin 120^\circ}{\sin 85.7^\circ}$$

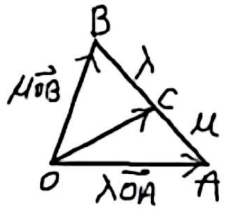
$$P = \frac{10 \times \sin 120^\circ}{2 \times \sin 85.7^\circ} \Rightarrow P = 4.35\,N$$

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Lecture No 20

Proof of Lambda mu theorem

(λ, μ) Theorem If two concurrent forces are represented by $\lambda\vec{OA}$ and $\mu\vec{OB}$, their resultant is given by $(\lambda+\mu)\vec{OC}$, where C divides AB such that $AC : CB = \mu : \lambda$



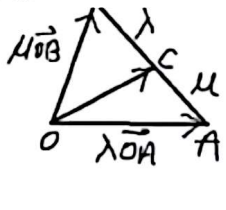
Proof $\vec{OA} = \vec{OC} + \vec{CA}$ —(1)

$\vec{OB} = \vec{OC} + \vec{CB}$ —(2)

(1)×λ $\lambda\vec{OA} = \lambda\vec{OC} + \lambda\vec{CA}$ —(3)

(2)×μ $\mu\vec{OB} = \mu\vec{OC} + \mu\vec{CB}$ —(4)

(3)+(4) $\lambda\vec{OA} + \mu\vec{OB} = \lambda\vec{OC} + \mu\vec{OC} + \lambda\vec{CA} + \mu\vec{CB}$ —(5)



$\vec{OA} = \vec{OC} + \vec{CA}$ —(1)

$\vec{OB} = \vec{OC} + \vec{CB}$ —(2)

(1)×λ $\lambda\vec{OA} = \lambda\vec{OC} + \lambda\vec{CA}$ —(3)

(2)×μ $\mu\vec{OB} = \mu\vec{OC} + \mu\vec{CB}$ —(4)

(3)+(4) $\lambda\vec{OA} + \mu\vec{OB} = \lambda\vec{OC} + \mu\vec{OC} + \lambda\vec{CA} + \mu\vec{CB}$ —(5)

$AC : CB = \mu : \lambda$ $\lambda\vec{AC} = \mu\vec{CB} \Rightarrow +\lambda\vec{CA} + \mu\vec{CB} = 0$

$\lambda\vec{OA} + \mu\vec{OB} = (\lambda+\mu)\vec{OC} - \lambda\vec{AC} + \mu\vec{CB}$

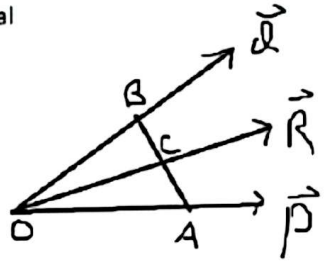
R :- $\lambda\vec{OA} + \mu\vec{OB} = (\lambda+\mu)\vec{OC}$ which is the proof.

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Example

Forces **P** and **Q** act at a point and theire resultant is **R.** If any transversal cuts three lines of action in the points A,B,C resp. prove that P/OA+Q/OB=R/OC



Proof

$$\vec{P} = P\,\hat{OA} = P\,\frac{\vec{OA}}{OA} = \frac{P}{OA}\,\vec{OA} \quad -(1)$$

$$\vec{Q} = Q\,\hat{OB} = Q\,\frac{\vec{OB}}{OB} = \frac{Q}{OB}\,\vec{OB} \quad -(2)$$

$$\vec{R} = R\,\hat{OC} = R\,\frac{\vec{OC}}{OC} = \frac{R}{OC}\,\vec{OC} \quad -(3)$$

$$\vec{R} = \vec{P} + \vec{Q}$$

$$= \frac{P}{OA}\,\vec{OA} + \frac{Q}{OB}\,\vec{OB}$$

$$\frac{R}{OC}\,\vec{OC} = \left(\frac{P}{OA} + \frac{Q}{OB}\right)\vec{R} = \left(\frac{P}{OA} + \frac{Q}{OB}\right)\vec{OC}$$

$$\frac{R}{OC} = \frac{P}{OA} + \frac{Q}{OB}$$

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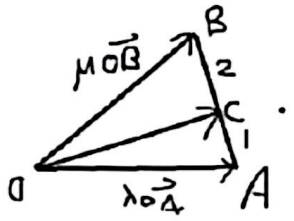
Example

Two concurrent forces are represented by lambdaOA and muOB.Find vector OC where C divides AB so that AC:CB=1:2 and OA=i+j-2k and OB=i+j+2k.

Sol:-

$\mu = 1, \lambda = 2$

$$\vec{R} = \lambda \vec{OA} + \mu \vec{OB} = (\lambda + \mu)\vec{OC}$$

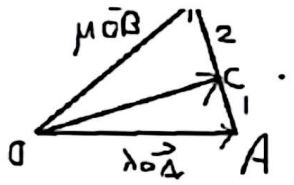


$$\lambda \vec{OA} + \mu \vec{OB} = (\lambda + \mu)\vec{OC}$$

$$2(i+j-2k) + 1(i+j+2k) = (2+1)\vec{OC}$$

$\mu = 1, \lambda = 2$

$$\vec{R} = \lambda \vec{OA} + \mu \vec{OB} = (\lambda + \mu)\vec{OC}$$



$$\lambda \vec{OA} + \mu \vec{OB} = (\lambda + \mu)\vec{OC}$$

$$2(i+j-2k) + 1(i+j+2k) = (2+1)\vec{OC}$$

$$2i + 2j - 4k + i + j + 2k = 3\vec{OC}$$

$$3i + 3j - 2k = 3\vec{OC}$$

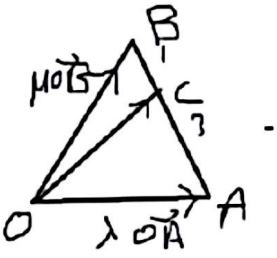
$$\vec{OC} = i + j - \frac{2}{3}k$$

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Lecture No 21

Example

Two concurrent forces are represented by lambdaOA and muOB. Find vector OA where C divides AB so that AC:CB=3:1 and OB=2i+j-5k and OC=4i+2j+k. Also find the resultant of these forces.



Sol:

$$\vec{R} = \lambda \overrightarrow{OA} + \mu \overrightarrow{OB} = (\lambda + \mu)\overrightarrow{OC} \quad \text{---(1)}$$

$\mu = 3, \lambda = 1$

by (1)

$$1\overrightarrow{OA} + 3(2i+j-5k) = (3+1)(4i+2j+k)$$

$$\vec{R} = \lambda \overrightarrow{OA} + \mu \overrightarrow{OB} = (\lambda + \mu)\overrightarrow{OC} \quad \text{---(1)}$$

$\mu = 3, \lambda = 1$

by (1)

$$1\overrightarrow{OA} + 3(2i+j-5k) = (3+1)(4i+2j+k)$$

$$\overrightarrow{OA} + 6i+3j-15k = 4(4i+2j+k)$$

$$\overrightarrow{OA} = 16i+8j+4k-6i-3j+15k$$

$$= 10i+5j+19k$$

Now

$$\vec{R} = (3+1)\overrightarrow{OC}$$

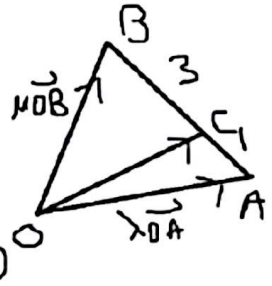
$$= 4(4i+2j+k)$$

$$= 16i+8j+4k$$

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Example

Two concurrent forces are represented by lambdaOAand muOB .Find vector OB where C divides AB so that AC:CB−1:3 and **OA=2i+3j+5k** and **OC=3i+3j+k.** Also find the resultant of these forces.



Sol..

By (λ-μ) Th.

$\vec{R} = \lambda\vec{OA} + \mu\vec{OB} = (\lambda+\mu)\vec{OC}$ —(1)

$\mu = 1, \quad \lambda = 3$

$3(2i+3j+5k) + \vec{OB} = (3+1)(3i+3j+k)$

$\mu = 1, \quad \lambda = 3$

$3(2i+3j+5k) + \vec{OB} = (3+1)(3i+3j+k)$

$6i+9j+15k + \vec{OB} = 4(3i+3j+k)$

$\vec{OB} = 12i + 12j + 4k - 6i - 9j - 15k$

$= 6i + 3j - 11k$

Now $\vec{R} = (\lambda+\mu)\vec{OC}$

$= (3+1)(3i+3j+k)$

$R = 12i + 12j + 4k$

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$\mu = 1, \quad \lambda = 3$

$$3(2i+3j+5k) + \vec{OB} = (3+1)(3i+3j+k)$$

$$6i+9j+15k + \vec{OB} = 4(3i+3j+k)$$

$$\vec{OB} = 12i + 12j + 4k - 6i - 9j - 15k$$

$$= 6i + 3j - 11k$$

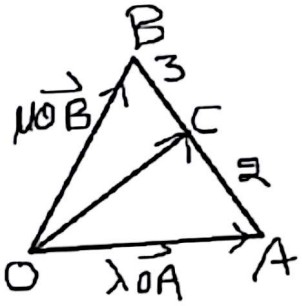
Now $\vec{R} = (\lambda + \mu)\vec{OC}$

$$= (3+1)(3i+3j+k)$$

$$\vec{R} = 12i + 12j + 4k$$

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Example
Two concurrent forces are represented by lambdaOAand muOB.Find the magnitude of the resultant force where C divides AB so that AC:CB−2:3 and **OA=i+3j+5k** and **OC=4i-j+k.** Also find the resultant of these forces.



Sol:-

By (λ-μ) Th

$\vec{R} = \lambda \vec{OA} + \mu \vec{OB} = (\lambda + \mu)\vec{OC}$

$\lambda = 3, \quad \mu = 2$

$\vec{R} = (\lambda + \mu)\vec{OC}$

$R = (\lambda + \mu) OC$

$= (3+2)(4i - j + k)$

$= 5(4i - j + k)$

$\vec{R} = 20i - 5j + 5k$

$R = \sqrt{20^2 + 5^2 + 5^2}$

$= \sqrt{400 + 25 + 25}$

$= \sqrt{450}$

$= \sqrt{9 \times 25 \times 2}$

$= 3 \times 5 \times \sqrt{2}$

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